

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-1:Lecture-3
Number Systems

CONTENTS

Module-1 (Part-1)

- ❖ **Number systems and conversions**
- ❖ **Binary arithmetic operations**
- ❖ r 's and $(r-1)$'s compliment
- ❖ Binary signed and unsigned numbers
- ❖ Weighted and non-weighted binary codes

CONTENTS

Lecture-3

- ❖ Conversion of different number systems (cont...)
- ❖ Binary Arithmetic

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

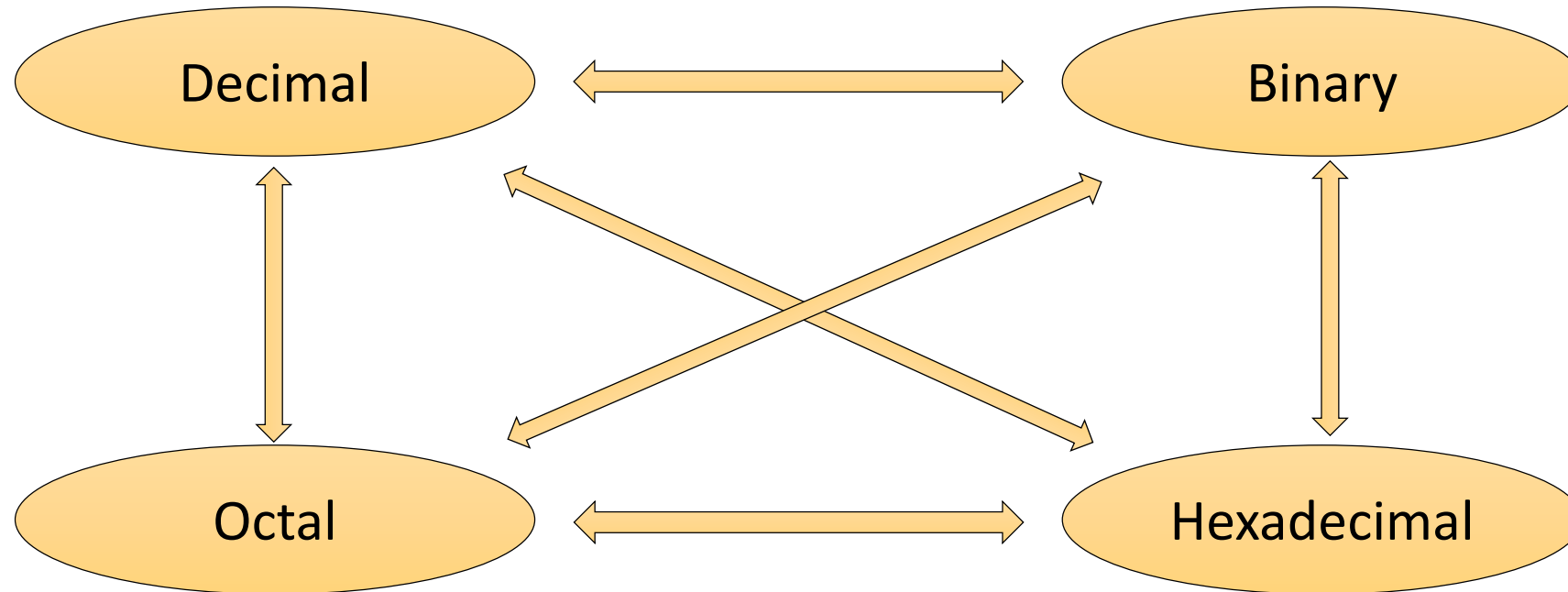
References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

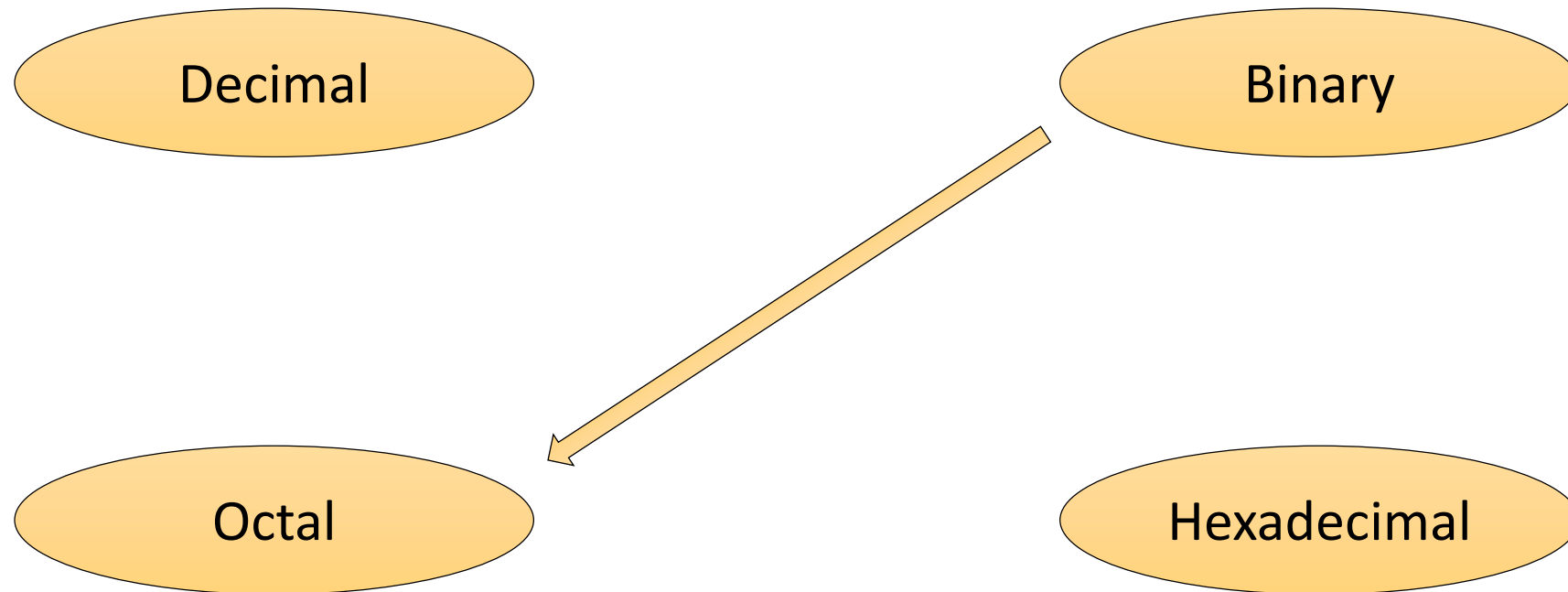
Number Systems

CONVERSION BETWEEN NUMBER BASES

✓ The possibilities:



CONVERSION FROM BINARY TO OCTAL

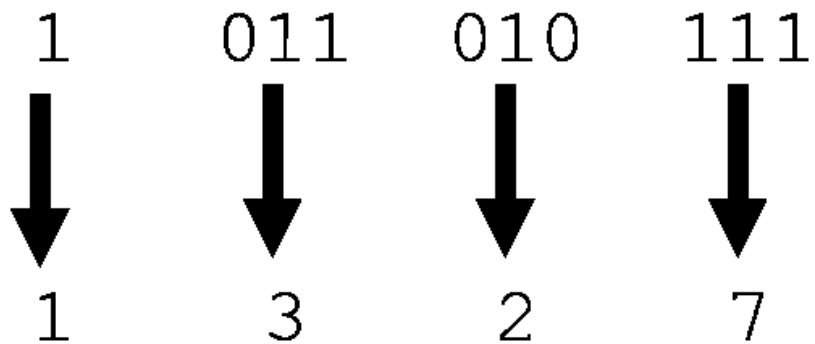


CONVERSION FROM BINARY TO OCTAL

✓ Technique:

1. Group bits in threes, starting on right for integer part and from left for fractional part.
2. Ignore leading zeros and add zeros after decimal if necessary, to form group of three bits.
3. Convert to octal digits

✓ Example: $(1011010111)_2 = (N)_8$



$$(1011010111)_2 = (1327)_8$$

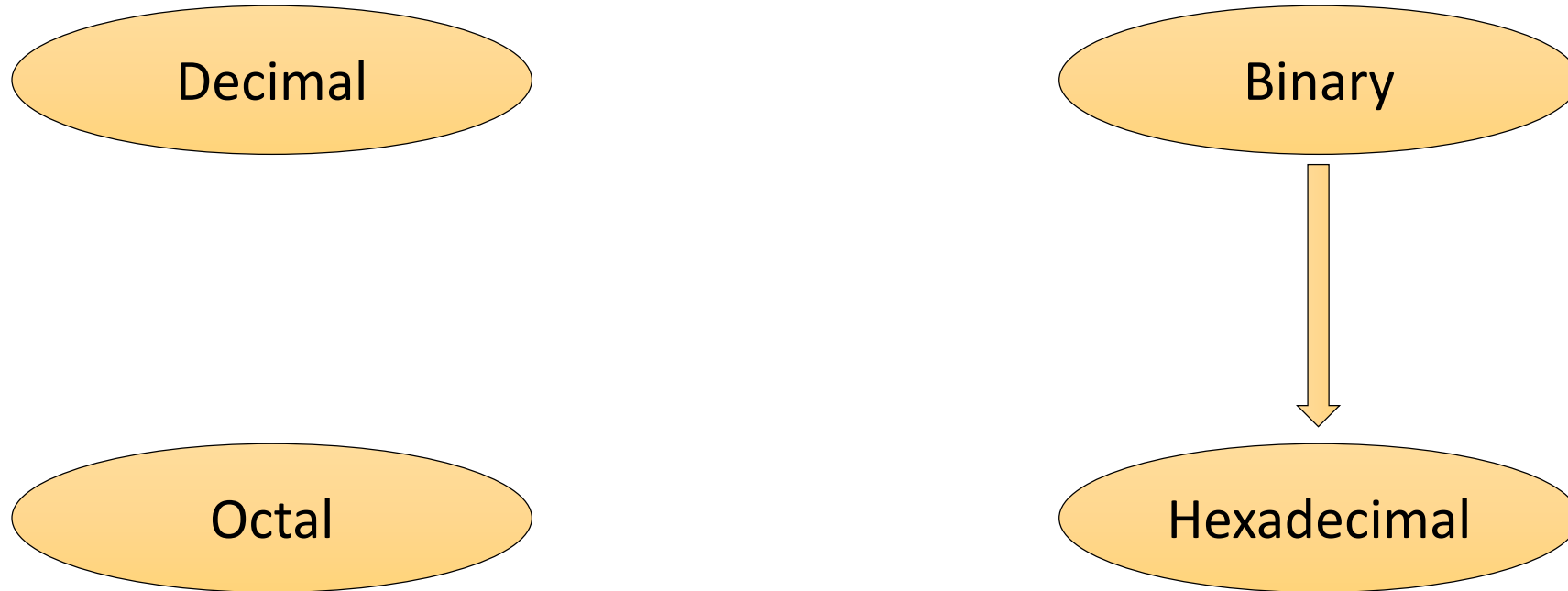
CONVERSION FROM BINARY TO OCTAL

✓ Example: $(1010111000.0101)_2 = (N)_8$

1	010	111	000	.	010	100
↓	↓	↓	↓		↓	↓
1	2	7	0	.	2	4

$$(1010111000.0101)_2 = (1270.24)_8$$

CONVERSION FROM BINARY TO HEXADECIMAL

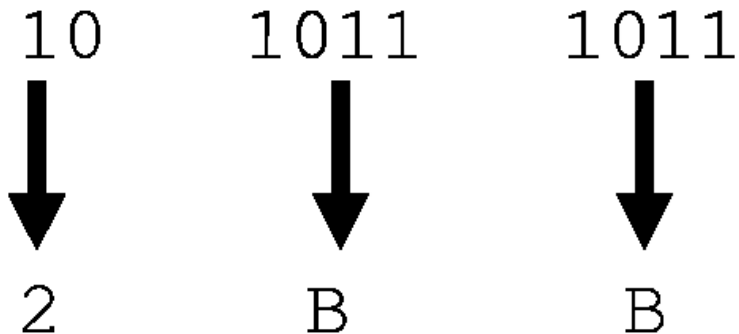


CONVERSION FROM BINARY TO HEXADECIMAL

✓ Technique:

1. Group bits in fours, starting on right for integer part and from left for fractional part.
2. Ignore leading zeros and add zeros after decimal if necessary, to form group of four bits.
3. Convert to hexadecimal digits

✓ Example: $(1010111011)_2 = (N)_{16}$



$$(1010111011)_2 = (2BB)_{16}$$

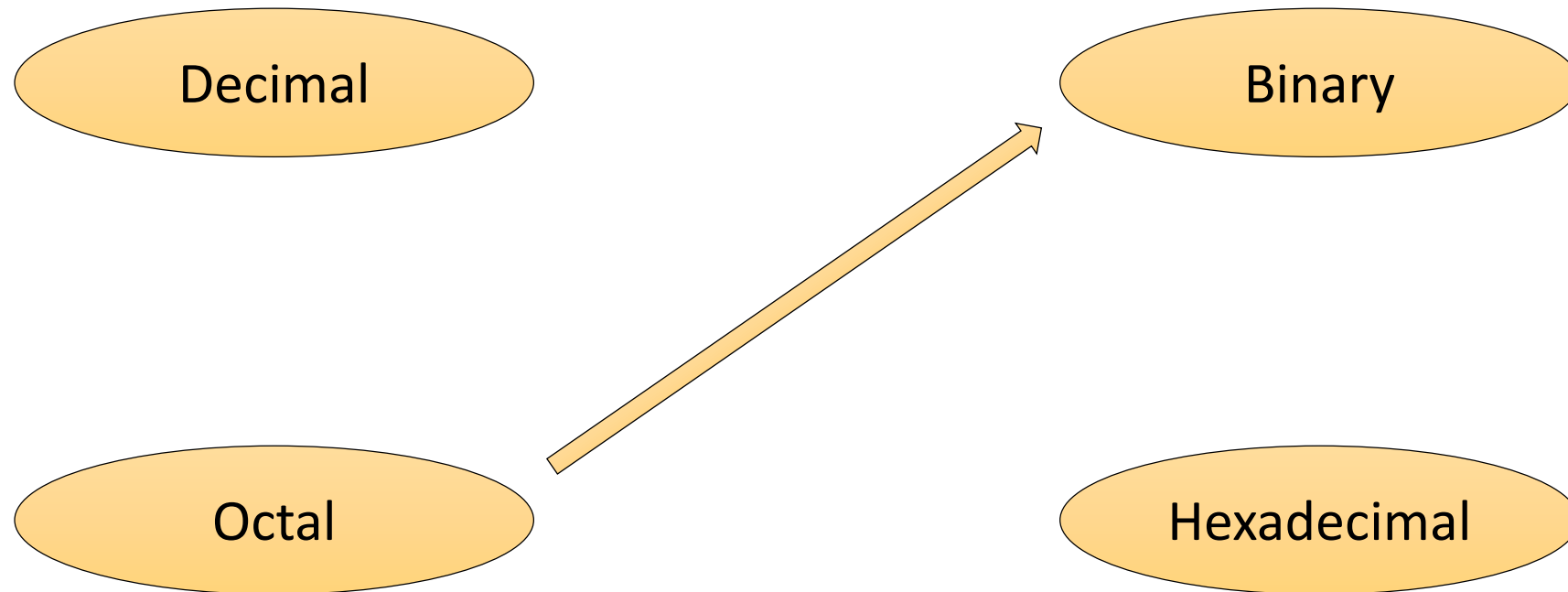
CONVERSION FROM BINARY TO OCTAL

✓ **Example:** $(1010111000.0101)_2 = (N)_{16}$

0010	1011	1000	.	0101
↓	↓	↓		↓
2	B	8	.	5

$$(1010111000.0101)_2 = (2B8.5)_{16}$$

CONVERSION FROM OCTAL TO BINARY

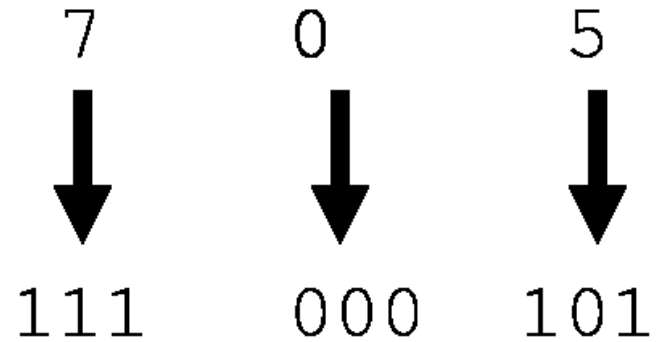


CONVERSION FROM OCTAL TO BINARY

✓ **Technique:**

1. Convert each octal digit to a 3-bit equivalent binary representation

✓ **Example:** $(705)_8 = (N)_2$



$$(705)_8 = (111000101)_2$$

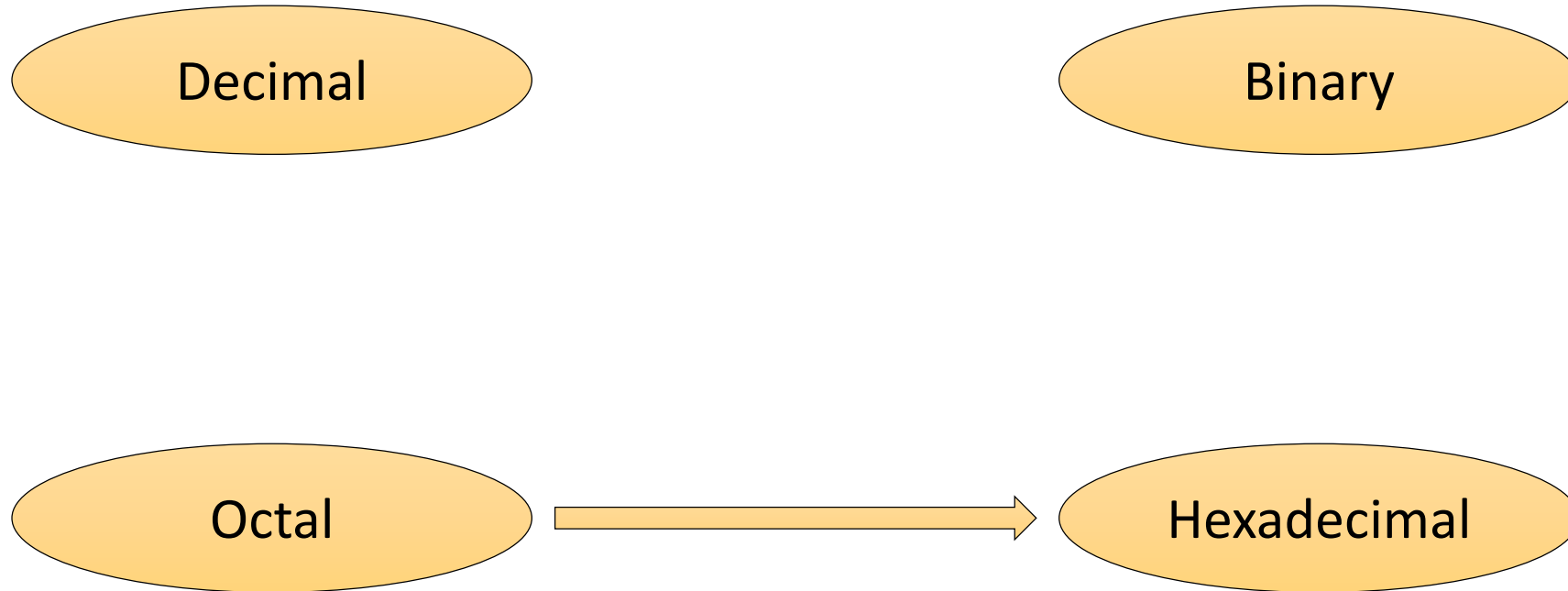
CONVERSION FROM OCTAL TO BINARY

✓ Example: $(1472.16)_8 = (N)_2$

1	4	7	2	.	1	6
↓	↓	↓	↓		↓	↓
1	100	111	010	.	001	110

$$(1472.16)_8 = (1100111010.00111)_2$$

CONVERSION FROM OCTAL TO HEXADECIMAL

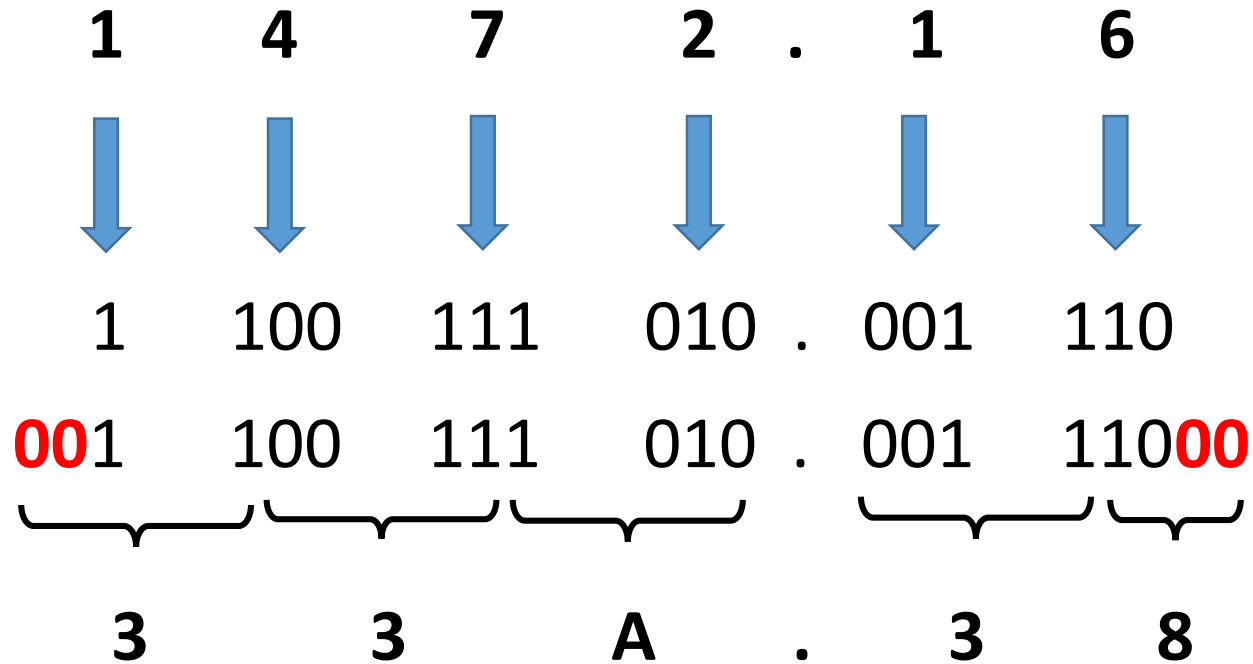


CONVERSION FROM OCTAL TO HEXADECIMAL

✓ Technique:

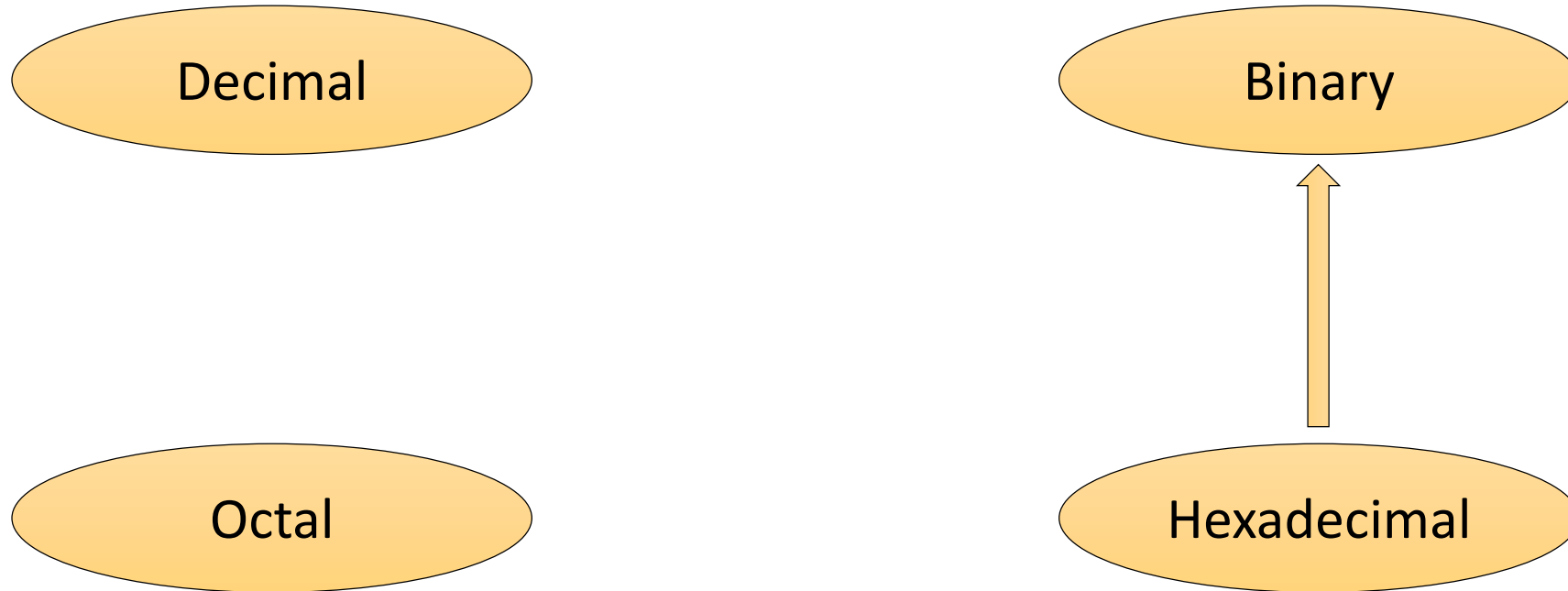
1. Use binary as an intermediary

✓ Example: $(1472.16)_8 = (N)_{16}$



$$(1472.16)_8 = (33A.38)_{16}$$

CONVERSION FROM HEXADECIMAL TO BINARY

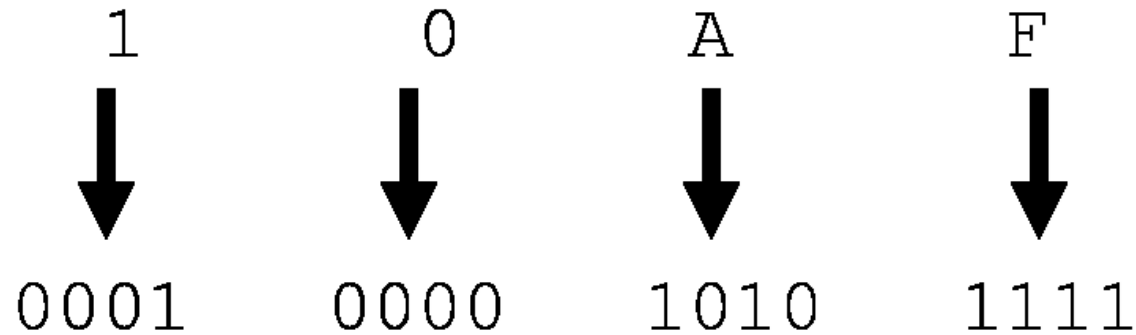


CONVERSION FROM HEXADECIMAL TO BINARY

✓ Technique:

1. Convert each hexadecimal digit to a 4-bit equivalent binary representation

✓ Example: $(10AF)_{16} = (N)_2$



$$(10AF)_{16} = (1000010101111)_2$$

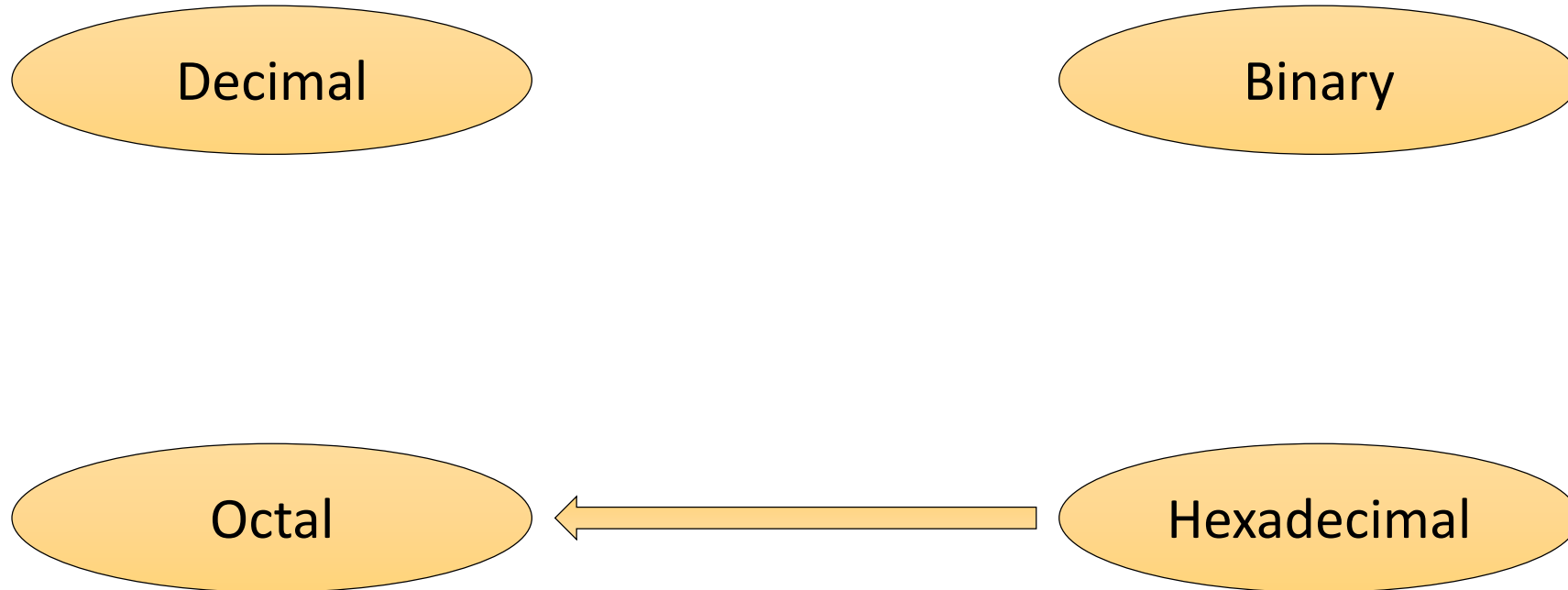
CONVERSION FROM HEXADECIMAL TO BINARY

✓ Example: $(A5BF.0C)_{16} = (N)_2$

A	5	B	F	.	0	C
↓	↓	↓	↓		↓	↓
1010	0101	1011	1111	.	0000	1100

$$(A5BF.0C)_{16} = (1010010110111111.000011)_2$$

CONVERSION FROM HEXADECIMAL TO OCTAL



CONVERSION FROM HEXADECIMAL TO OCTAL

✓ Technique:

1. Use binary as an intermediary

✓ Example: $(A5BF.0C)_{16} = (N)_8$

A	5	B	F	.	0	C
↓	↓	↓	↓		↓	↓
1010	0101	1011	1111	.	0000	1100
001010	0101	1011	1111	.	0000	1100
1 2	2 6	7 7		.	0	3

$$(A5BF.0C)_{16} = (122677.03)_8$$

ASSIGNMENT QUESTION

✓ Fill the table:

Decimal	Binary	Octal	Hexa- decimal
29.8			
	101.1101		
		3.07	
			C.82

Binary Arithmetic

BINARY ADDITION

✓ Two 1-bit values:

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	10

"two"

BINARY ADDITION

✓ Two n-bit values:

- Add individual bits
- Propagate carries

✓ Example:

$$\begin{array}{rcccccc} & & 1 & 1 & 1 & 1 & 1 & 1 & \leftarrow \text{carries} \\ & & 1 & 1 & 1 & 1 & 0 & 1 \\ + & & & 1 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{r} 6 & 1 \\ + & 2 & 3 \\ \hline 8 & 4 \end{array}$$

BINARY ADDITION

✓ **Two n-bit values:**

- Add individual bits
- Propagate carries

✓ **Example:**

$$\begin{array}{r} \overset{1}{1}01\overset{1}{0}1 \\ + 11001 \\ \hline 101110 \end{array}$$

$$\begin{array}{r} 21 \\ + 25 \\ \hline 46 \end{array}$$

BINARY SUBTRACTION

✓ Two 1-bit values:

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	0

With Borrow 1

BINARY SUBTRACTION

✓ Two n-bit values:

- Subtract individual bits with borrow if necessary

✓ Example:

$$\begin{array}{r} 111101 \\ + 101111 \\ \hline 100110 \end{array}$$

Diagram illustrating binary subtraction with borrows. The top number is 111101 and the bottom number is 101111. The result is 100110. Red annotations show the borrowing process: a red '10' is written above the third bit from the right, with an arrow pointing from the fourth bit to it. Another red '10' is written above the second bit from the right, with an arrow pointing from the third bit to it. A third red '10' is written above the first bit from the right, with an arrow pointing from the second bit to it. A label 'Borrows' with an arrow points to the third red '10'.

10- decimal value 2

$$\begin{array}{r} 61 \\ - 23 \\ \hline 38 \end{array}$$

BINARY MULTIPLICATION

✓ Two 1-bit values:

A	B	$A \times B$
0	0	0
0	1	0
1	0	0
1	1	1

BINARY MULTIPLICATION

✓ **Example:**

				1	0	1	1	1
x					1	0	1	0

				0	0	0	0	0
		1		0	1	1	1	
	0	0		0	0	0		
1	0	1		1	1			

	1	1		0	0	1	1	0

	2	3
x	1	0
	2	3
	0	0

ASSIGNMENT QUESTIONS

1. Add: $(1000111)_2 + (1110)_2$
2. Subtract: $(1110000)_2 - (1111)_2$
3. Multiply: $(101)_2 \times (10101)_2$

Thank You